Permit No. MI0020257

FACT SHEET

PERMITTEE/FACILITY NAME: City of Cadillac / Cadillac WWTP

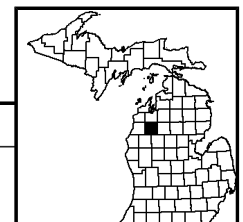
COUNTY: WEXFORD

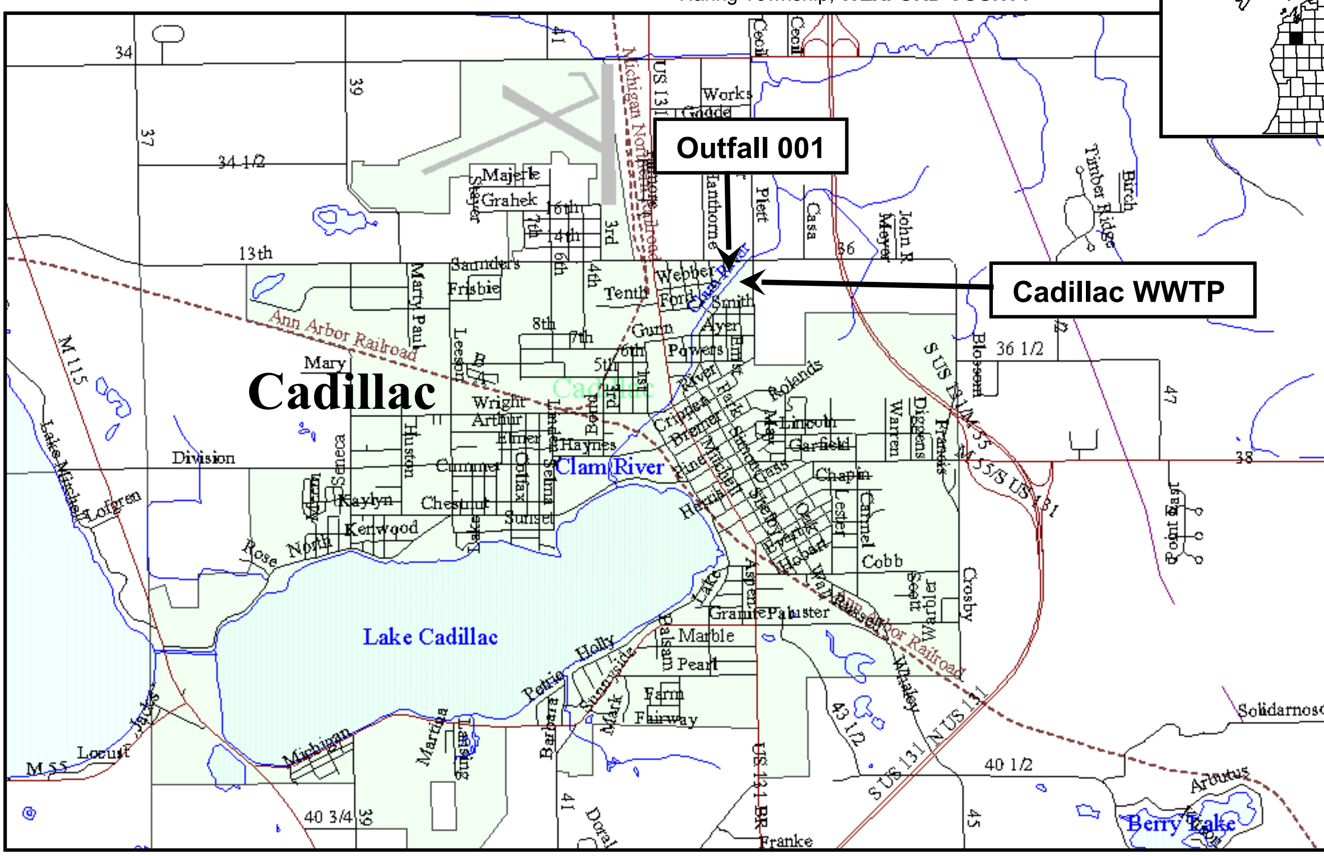
DESCRIPTION OF EXISTING WASTEWATER TREATMENT FACILITIES

The Cadillac Wastewater Treatment Plant (WWTP) was originally constructed in 1963 as a secondary treatment activated sludge facility. Improvements in 1975, 1990, and 1996 added Rotating Biological Contactors and fine bubble diffusers to improve biological treatment and ammonia removal; additional secondary clarifiers and sand filters for solids removal; and further improvements in equalization, grit and solids handling, and biosolids storage. In 2003 the sodium hypochlorite disinfection system was replaced with an ultraviolet (UV) light disinfection system. Phosphorus is removed with ferric chloride. The sand filters have been replaced with Aqua Disk cloth media. Sludge generated at the facility is stabilized in an anaerobic digester and stored until land applied. Treated effluent and storm water collected from the facility grounds are discharged to the Clam River.

MAP OF DISCHARGE LOCATION

Facility Location Coordinates:
NE1/4, NE1/4, Section 33, Town 22N, Range 9 W
Haring Township, **WEXFORD COUNTY**





RECEIVING WATER

The Clam River is protected for agricultural uses, navigation, industrial water supply, public water supply in areas with designated public water supply intakes, warmwater fish from Lake Cadillac to County Line Road and coldwater fish downstream of County Line Road, other indigenous aquatic life and wildlife, partial body contact recreation, total body contact recreation (May through October), and fish consumption.

The receiving stream flows used to develop effluent limitations are a 95 percent exceedance flow of 2.7 cfs, a harmonic mean flow of 13 cfs, and a 90-day, 10-year low flow of 3.3 cfs.

MIXING ZONE

For toxic pollutants, the volume of the Clam River used in assuring that effluent limitations are sufficiently stringent to meet Water Quality Standards is 25 percent of the applicable design flow of the receiving stream.

For other pollutants, the volume of the Clam River used in assuring that effluent limitations are sufficiently stringent to meet Water Quality Standards is the applicable design flow of the receiving stream.

EXISTING EFFLUENT QUALITY: (from DMR data from 1-1-2010 to 3-31-2013)

Parameter	Minimum Daily	Maximum Monthly	Maximum 7-day	Maximum Daily	Units
Flow	---	2.668	---	3.752	MGD
Carbonaceous Biochemical Oxygen Demand ($CBOD_5$)					
May 1 – Nov. 30	---	4.1	---	13.3	mg/l
Dec. 1 – Mar. 31	---	4.9	---	13.4	mg/l
Apr. 1 – Apr. 30	---	3.0	---	5.9	mg/l
Total Suspended Solids	93.33	---	---	---	% Removal
May 1 – Nov. 30	---	3.8	3.9	---	mg/l
Dec. 1 – Apr. 30	---	5.3	5.6	---	mg/l
Ammonia Nitrogen (as N)					
May 1 – Nov. 30	---	0.6	---	1.9	mg/l
Dec. 1 – Mar. 31	---	0.7	---	---	mg/l
Apr. 1 – Apr. 30	---	0.4	---	---	mg/l
Total Phosphorus (as P)					
May 1 – Mar. 31	---	0.4	---	---	mg/l
Apr. 1 – Apr. 30	---	0.19	---	---	mg/l
Fecal Coliform Bacteria	---	111	475	---	cts/100ml
Total Mercury	---	0.8	---	---	ng/l
Acute Toxicity	---	---	---	0	TU_A
Chronic Toxicity	---	2.4	---	---	TU_C
Total Zinc	---	---	---	167	µg/l
Total Selenium	---	---	---	2.0	µg/l
Available Cyanide	---	---	---	2.0	µg/l
pH	6.4	---	---	8.0	S.U.

Dissolved Oxygen					
May 1 – Nov. 30	7.1	---	---	---	mg/l
Dec. 1 – Apr. 30	7.8	---	---	---	mg/l

PROPOSED EFFLUENT LIMITATIONS: (see draft permit)

BASIS FOR PROPOSED EFFLUENT LIMITATIONS (see Basis for Decision memo)

ADDITIONAL INFORMATION

This permit requires the permittee to prepare an Asset Management Program, and implement the program upon Department approval. Annual Reports summarizing program activity are also required to be submitted. The requirements of an Asset Management Program contain goals of effective performance, adequate funding, and adequate operator staffing and training. Asset management is a planning process that ensures gaining optimum value for each asset and providing the financial resources to rehabilitate and replace them when necessary; and typically includes five core elements which identify: the current state of the asset, the desired level of service (e.g., per the permit, or for the customer), the most critical asset(s) to sustain performance, the best life cycle cost, and the long term funding strategy to sustain service and performance.

REGISTER OF INTERESTED PERSONS

Any person interested in a particular application, or group of applications, may leave his/her name, address, and telephone number as part of the file for an application. The list of names will be maintained as a means for persons with an interest in an application to contact others with similar interests.

PUBLIC COMMENT

Comments or objections to the draft permit received between August 20, 2013, and September 19, 2013, will be considered in the final decision to issue the permit.

If submitted comments indicate significant public interest in the application or if useful information may be produced, the Department, at its discretion, may hold a public hearing on the application. Any person may request the Department to hold a public hearing on the application. The request should include specific reasons for the request, indicating which portions of the application or draft permit constitute the need for a hearing.

Public notice of a hearing will be provided at least thirty (30) days in advance. The hearing will normally be held in the vicinity of the discharge. The Department will consider comments made at the hearing when making its final determinations on the permit. Further information regarding the draft permit, and procedures for commenting or requesting a public hearing may be obtained by contacting Mike Bitondo, Permits Section, Water Resources Division, Department of Environmental Quality, P.O. Box 30458, Lansing, Michigan 48909, telephone: 517-335-3303, e-mail: bitondom@michigan.gov.